

SOLUTION DESIGN AND IMPLEMENTATION

1. System Architecture

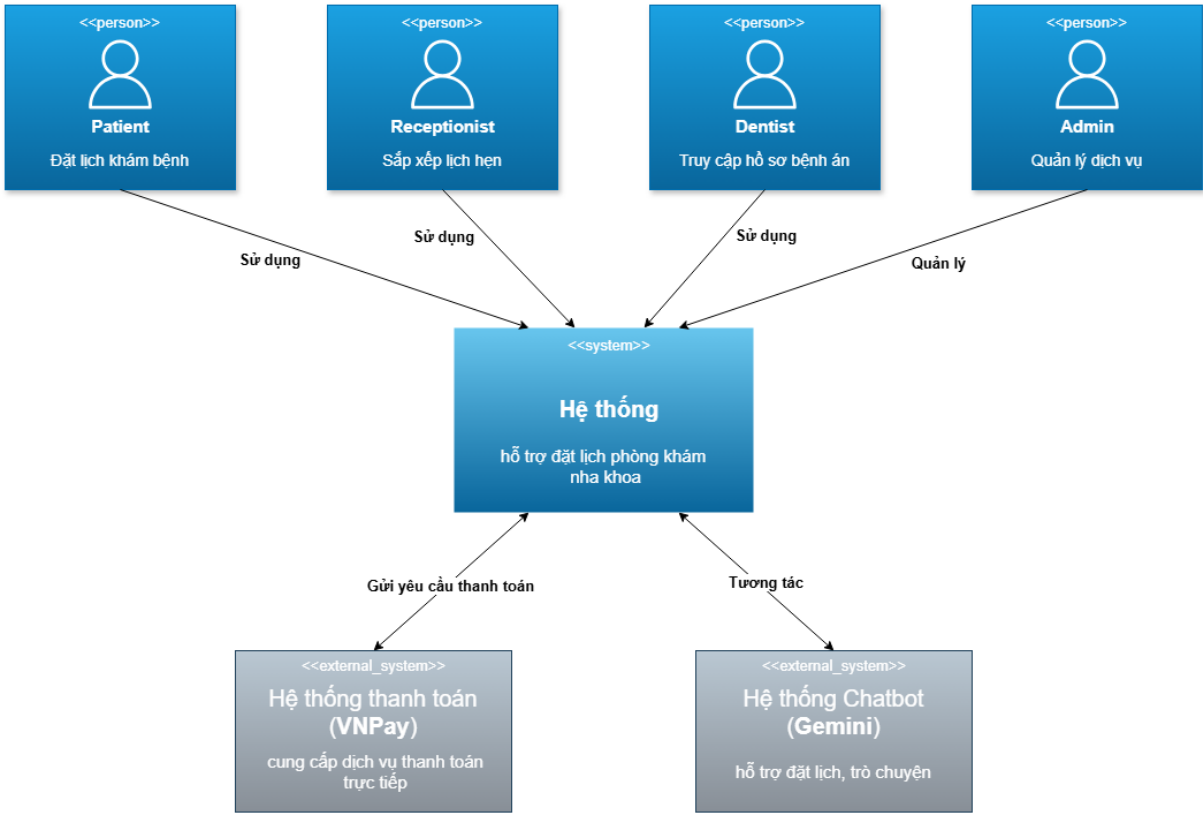


Figure 1. Context Diagram

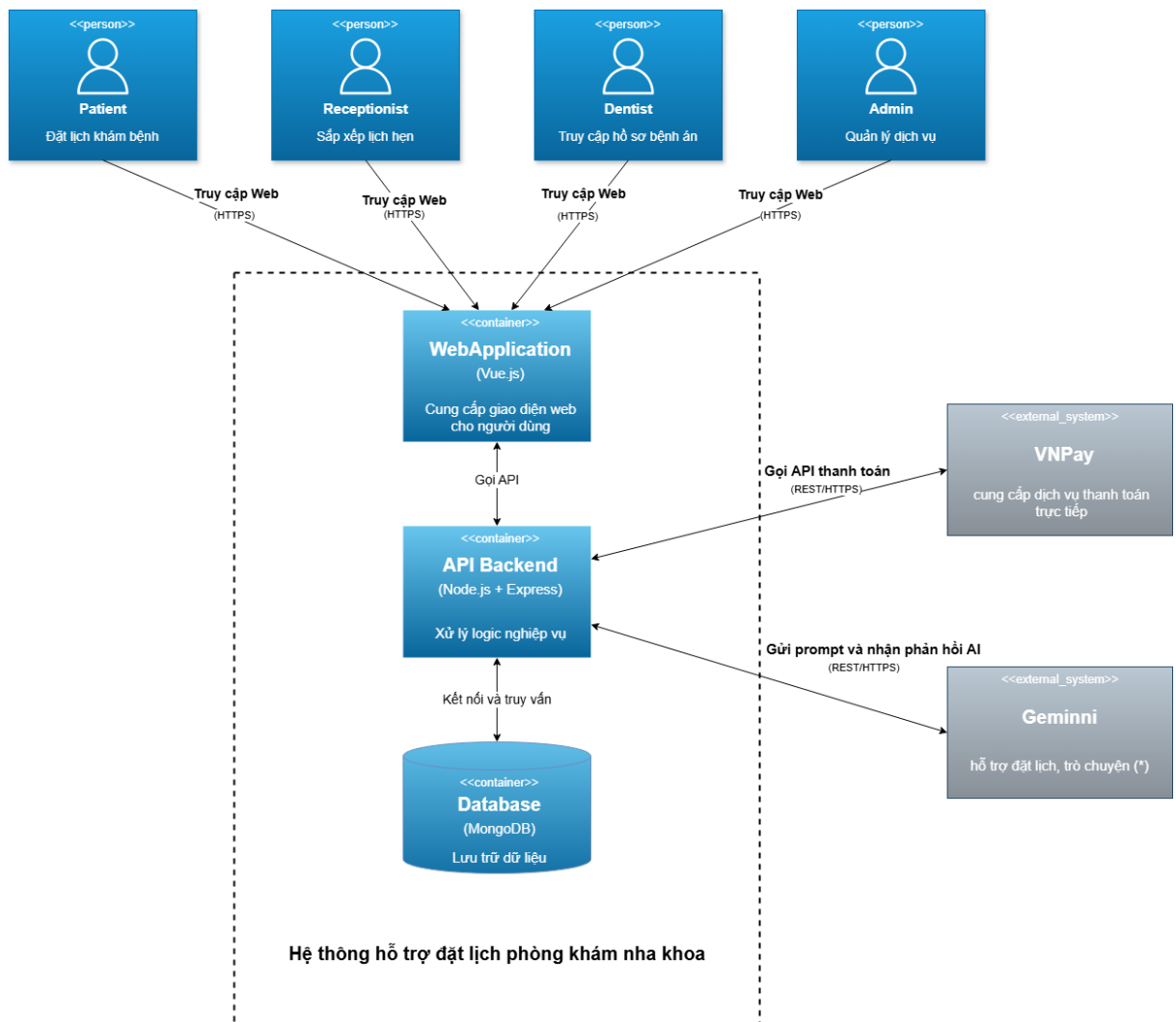


Figure 2. Container Diagram

2. Data Design

2.1. Data Model

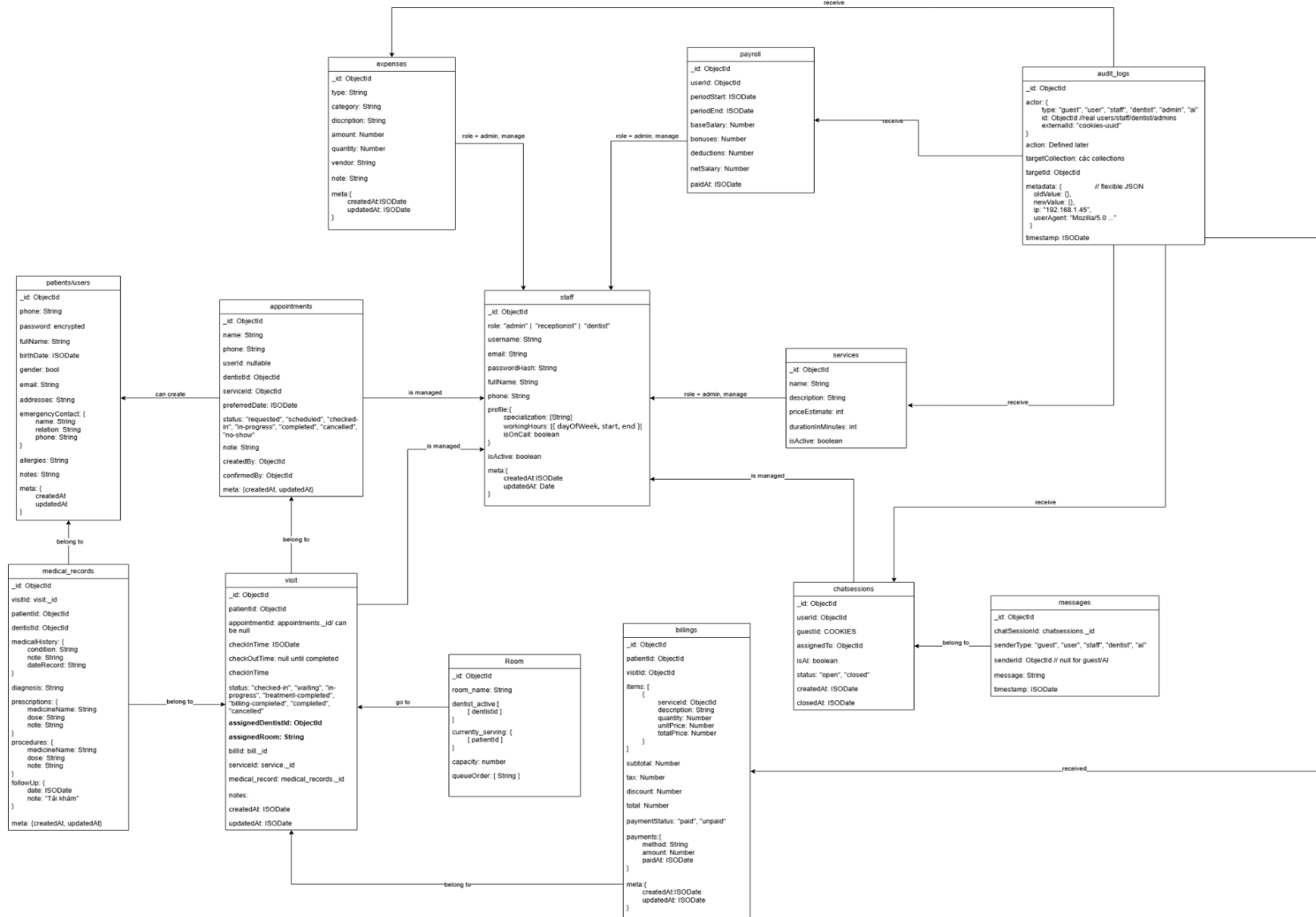


Figure 4. Physical Data Modeling

2.2. Data Dictionary

2.2.1. Collection appointments

Stores information about patient schedules, service details, and payment records, serving the mechanisms for appointment management, workflow tracking, and billing processing.

Table 1. Collection appointments

No	Field name	Data type	Nullable	PK/FK	Constraint	Description
1	_id	ObjectId	No	PK	Unique	Unique identifier for the appointment.
2	patientId	ObjectId	No	FK	Ref: Patients	Reference ID linking to the patient profile.
3	patient	String	Yes	-	-	Cached name of the patient.
4	dentistId	ObjectId	Yes	FK	Ref: Dentists	Reference ID linking to the assigned dentist.
5	dentist	String	Yes	-	-	Cached name or details of the dentist.
6	receptionistId	ObjectId	Yes	FK	Ref: Staff	Reference ID of the receptionist managing the record.
7	date	String	No	-	Format: YYYY-MM-DD	The scheduled date of the appointment.
8	time	String	No	-	Format: HH:MM AM/PM	The scheduled time slot.
9	status	String	No	-	Enum: ["Requested", "Scheduled", "Done", "Cancelled"]	Current processing status of the appointment.
10	serviceId	String	No	FK	Ref: Services	Unique code/ID of the selected dental service.

11	service	String	Yes	-	-	Name of the dental service.
12	reason	String	Yes	-	-	Symptom or reason for the visit provided by the patient.
13	phone	String	No	-	-	Patient's contact phone number.
14	requestId	String	No	-	Unique	Human-readable ID for tracking requests.
15	notes	String	Yes	-	-	Additional administrative or medical notes.
16	completedBy	ObjectId	Yes	FK	Ref: Staff	ID of the staff who marked the appointment as done.
17	paymentProcessedBy	ObjectId	Yes	FK	Ref: Staff	ID of the staff who processed the payment.
18	paymentAmount	Number	Yes	-	Min: 0	Total cost charged for the appointment.
19	paymentType	String	Yes	-	Enum: ["Direct", "Insurance"]	Category of the payment transaction.
20	paymentMethod	String	Yes	-	Enum: ["Cash", "Card"]	Method used for payment.
21	paymentDate	String	Yes	-	Format: YYYY-MM-DD	Date when the payment was finalized.
22	createdAt	Date	No	-	-	Timestamp of record creation.
23	updatedAt	Date	No	-	-	Timestamp of the last modification.

2.2.2. Collection bills

Stores invoice information and transaction details for dental services, serving the mechanisms for payment processing, revenue tracking, and financial auditing.

Table 2. Collection bills

No	Field name	Data type	Nullable	PK/FK	Constraint	Description
1	_id	ObjectId	No	PK	Unique	Unique identifier for the bill/invoice.
2	appointmentId	ObjectId	No	FK	Ref: Appointments	Reference ID linking to the specific appointment.
3	patientId	ObjectId	No	FK	Ref: Patients	Reference ID linking to the patient profile.
4	invoiceNumber	String	No	-	Unique	Human-readable unique invoice code.
5	service	String	Yes	-	-	Name of the primary service rendered.
6	date	String	No	-	Format: YYYY-MM-DD	The date the invoice was issued.
7	dentist	ObjectId/String	Yes	FK	Ref: Dentists	Reference to the dentist performing the service (nullable).
8	amount	Number	No	-	Min: 0	Total monetary amount for the bill.
9	status	String	No	-	Enum: ["Unpaid", "Paid", "Pending"]	Current payment status of the bill.
10	items	Array	Yes	-	-	List of specific line items (treatments/products) included in the bill.
11	paymentMethod	String	Yes	-	Enum: ["Cash", "Card", "Transfer"]	The method used by the patient to pay.
12	lastPaymentDate	String	Yes	-	Format:	The date when the payment was last processed or

					YYYY-MM-DD	completed.
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2.2.3. Collection chats

Manages the lifecycle and metadata of communication sessions between patients and the system, tracking the duration, status, and type of interaction for monitoring and history purposes.

Table 3. Collection chats

No	Field name	Data type	Nullable	PK/FK	Constraint	Description
1	_id	ObjectId	No	PK	Unique	Unique identifier for the chat session.
2	patientId	ObjectId	No	FK	Ref: Patients	Reference ID linking to the patient involved in the chat.
3	startTime	Date/String	No	-	-	Timestamp when the chat session was initiated.
4	endTime	Date/String	Yes	-	-	Timestamp when the chat session was closed or resolved.
5	status	String	No	-	Enum: ["open", "closed", "pending"]	Current operational status of the session.
6	chatType	String	No	-	Enum: ["normal", "emergency", "bot", "agent"]	Classification of the chat session type.

2.2.4. Collection chatmessages

Stores the individual text messages exchanged within a specific chat session, recording the sender's role, content, and timing to maintain the full conversation history and context for AI analysis or staff review.

Table 4. Collection chatmessages

No	Field name	Data type	Nullable	PK/FK	Constraint	Description
1	_id	ObjectId	No	PK	Unique	Unique identifier for the specific message.
2	chatId	ObjectId	No	FK	Ref: Chats	Reference ID linking the message to a specific chat session.
3	role	String	No	-	Enum: ["patient", "bot", "staff", "system"]	Identifies who sent the message (Sender).
4	messageText	String	No	-	-	The actual textual content of the message.
5	timestamp	Date/String	No	-	-	Precise timestamp when the message was sent.

2.2.5. Collection clinicsetting

Stores global configuration parameters for the system, including operating hours, clinic contact details, and notification templates, serving as the central reference for business rules and application customization.

Table 5. Collection clinicsetting

No	Field name	Data type	Nullable	PK/FK	Constraint	Description
1	_id	ObjectId	No	PK	Unique	Unique identifier for the settings document.
2	workingHours	Object	No	-	-	Structure defining opening and closing times for each day of the week.
3	holidays	Array	Yes	-	-	List of specific dates when the clinic is closed.

4	clinicInfo	Object	No	-	-	Object containing clinic details (name, address, phone, email).
5	notificationTemplates	Object	No	-	-	Predefined text templates used for automated system notifications.
6	timeSlotInterval	Number	No	-	Min: 15	The duration (in minutes) for each standard appointment slot.
7	updatedAt	ObjectId	Yes	FK	Ref: Staff	Reference ID of the administrator who last modified the settings.
8	createdAt	Date/String	No	-	-	Timestamp when the settings were initially created.
9	updatedAt	Date/String	No	-	-	Timestamp when the settings were last updated.

2.2.6. Collection inventories

Stores information about dental supplies and medical consumables, serving the mechanisms for stock tracking, supply chain management, and cost control within the clinic.

Table 6. Collection inventories

No	Field name	Data type	Nullable	PK/FK	Constraint	Description
1	_id	ObjectId	No	PK	Unique	Unique identifier for the inventory item.
2	name	String	No	-	-	Name of the supply or equipment item.
3	category	String	No	-	Enum: ["Consumables", "Instruments", etc.]	Classification category of the item.
4	quantity	Number	No	-	Min: 0	Current quantity available in stock.
5	unit	String	No	-	-	Unit of measurement (e.g., box, piece, bottle).

6	minStockLevel	Number	No	-	Min: 0	Minimum threshold to trigger a restock alert.
7	maxStockLevel	Number	Yes	-	Min: 0	Maximum desired quantity to prevent overstocking.
8	supplier	String	Yes	-	-	Name or identifier of the item's supplier.
9	costPerUnit	Number	Yes	-	Min: 0	Cost price per single unit of the item.
10	lastRestocked	Date/String	Yes	-	-	The date when the item was last replenished.
11	isActive	Boolean	No	-	Default: true	Flag to indicate if the item is currently in use/active.
12	createdAt	Date/String	No	-	-	Timestamp when the item was added to inventory.

2.2.7. Collection medicalrecords

Stores the comprehensive clinical history and health profile of patients, including treatment logs, prescriptions, allergies, and visit summaries, serving as the core reference for dentists to monitor patient health and ensure continuity of care.

Table 7. Collection medicalrecords

No	Field name	Data type	Nullable	PK/FK	Constraint	Description
1	_id	ObjectId	No	PK	Unique	Unique identifier for the medical record document.
2	patientId	ObjectId	No	FK	Ref: Patients	Reference ID linking to the specific patient.
3	patient	String	Yes	-	-	Cached name of the patient for quick display.
4	patientIdNumber	String	Yes	-	-	Citizen ID or Social Security Number of the patient.
5	lastVisit	Date/String	Yes	-	-	The date of the patient's most recent visit.
6	primaryDentist	String/ObjectId	Yes	-	-	Name or ID of the main dentist responsible for this patient.
7	totalVisits	Number	No	-	Min: 0	Counter tracking the total number of visits made by the patient.

8	allergies	String/Array	Yes	-	-	Known medical allergies of the patient (e.g., Penicillin).
9	treatments	Array	Yes	-	-	History list of treatments performed on the patient.
10	notes	Array	Yes	-	-	List of clinical observations or administrative notes.
11	prescriptions	Array	Yes	-	-	History list of medications prescribed to the patient.
12	reports	Array	Yes	-	-	List of attached medical files, X-rays, or lab reports.
13	visits	Array	Yes	FK	Ref: Appointments	List of references to past appointment IDs.

2.2.8. Collection procedures

Maintains the master catalog of dental services offered by the clinic, defining standard pricing, duration, and categories, serving as the foundational reference for appointment scheduling and invoice generation.

Table 8. Collection procedures

No	Field name	Data type	Nullable	PK/FK	Constraint	Description
1	_id	ObjectId	No	PK	Unique	Unique identifier for the procedure.
2	code	String	No	-	Unique	Short code or SKU identifying the service (e.g., "CLEANING").
3	name	String	No	-	-	Full display name of the dental service.
4	category	String	No	-	-	Classification/Type of the procedure (e.g., Preventive, Cosmetic).
5	duration	Number	No	-	Min: 0	Estimated time required for the procedure (in minutes).
6	price	Number	No	-	Min: 0	Standard cost/price of the service.
7	description	String	Yes	-	-	Detailed explanation of what the procedure involves.

8	isActive	Boolean	No	-	Default: true	Flag indicating if the service is currently offered.
9	createdAt	Date/String	No	-	-	Timestamp when the procedure was added to the catalog.

2.2.9. Collection treatmentplans

Stores detailed proposals for patient care, outlining specific procedures, estimated costs, and durations, serving as a roadmap for clinical execution and financial agreement between the clinic and the patient.

Table 9. Collection treatmentplans

No	Field name	Data type	Nullable	PK/FK	Constraint	Description
1	_id	ObjectId	No	PK	Unique	_id
2	patientId	ObjectId	No	FK	Ref: Patients	patientId
3	patientName	String	Yes	-	-	patientName
4	appointmentId	ObjectId	Yes	FK	Ref: Appointments	appointmentId
5	dentistId	ObjectId	No	FK	Ref: Dentists	dentistId
6	dentistName	String	Yes	-	-	dentistName
7	status	String	No	-	Enum: ["draft", "pending", "approved", "completed"]	status
8	description	String	Yes	-	-	description
9	procedures	Array	No	-	Min Items: 1	procedures

10	estimatedCost	Number	No	-	Min: 0	estimatedCost
11	estimatedDuration	Number	Yes	-	Min: 0	estimatedDuration
12	approvedBy	ObjectId	Yes	FK	Ref: Staff	approvedBy
13	approvedAt	Date/String	Yes	-	-	approvedAt
14	createdAt	Date/String	No	-	-	createdAt
15	updatedAt	Date/String	No	-	-	updatedAt

2.2.10. Collection users

Serves as the centralized repository for all system accounts (Admins, Dentists, Receptionists, and Patients), managing authentication credentials, personal profiles, and role-based access control (RBAC) to ensure secure system operation.

Table 10. Collection users

No	Field name	Data type	Nullable	PK/FK	Constraint	Description
1	_id	ObjectId	No	PK	Unique	_id
2	username	String	No	-	Unique	username
3	password	String	No	-	Hashed (Bcrypt)	password
4	role	String	No	-	Enum: ["admin", "dentist", "receptionist", "patient"]	role
5	name	String	No	-	-	name
6	email	String	No	-	Unique,	email

					Email Format	
7	phone	String	Yes	-	-	phone
8	address	String	Yes	-	-	address
9	dateOfBirth	Date/String	Yes	-	-	dateOfBirth
10	specialty	String	Yes	-	-	specialty
11	licenseNumber	String	Yes	-	Unique	licenseNumber
12	isActive	Boolean	No	-	Default: true	isActive
13	createdAt	Date/String	No	-	-	createdAt

3. Functional Design